

AUTO DATA STORYTELLER REPORT

Executive Summary

****Executive Summary**** Our analysis has yielded a robust predictive capability that demonstrates strong performance, reliably classifying outcomes with an accuracy exceeding 83%. This provides a highly dependable foundation for data-driven decision-making, offering valuable insights to inform strategic planning and operational adjustments. The reliability of these insights means we can confidently act on the identified patterns and drivers. The primary factors influencing these predictions are gender, the price paid, and age. Gender, specifically, emerges as the most significant determinant of the outcome, followed closely by the cost associated with the transaction and the individual's age. This highlights the critical role that both demographic profiles and transactional values play in shaping business outcomes, suggesting specific areas for focus in strategy and targeting. Given the strong predictive power and the clear identification of these influential factors, we recommend continuous monitoring of trends related to gender, pricing strategies, and age demographics. Changes within these key variables are likely to have the most substantial impact on overall outcomes. Leveraging these insights will enable more effective strategic planning and optimized resource allocation by focusing on the most impactful aspects of your operational or customer landscape.

Dataset Overview

Problem Type: classification

Best Model: Random Forest

Data Quality

- Dataset contains 891 rows & 12 columns
- 866 missing values were identified.
- Column Cabin has the highest missing values (687 values)
- No strong relationships were identified.

Model Findings

CLASSIFICATION MODEL PERFORMANCE

- Random Forest was selected as the best performing model based on F1.
- Random Forest achieved the highest F1 score of 0.83.
- The model correctly classified approximately 83.2% of records.
- The model demonstrates strong classification performance.
- Random Forest outperformed Logistic Regression, Decision Tree based on F1.

Key Drivers

MODEL DRIVERS

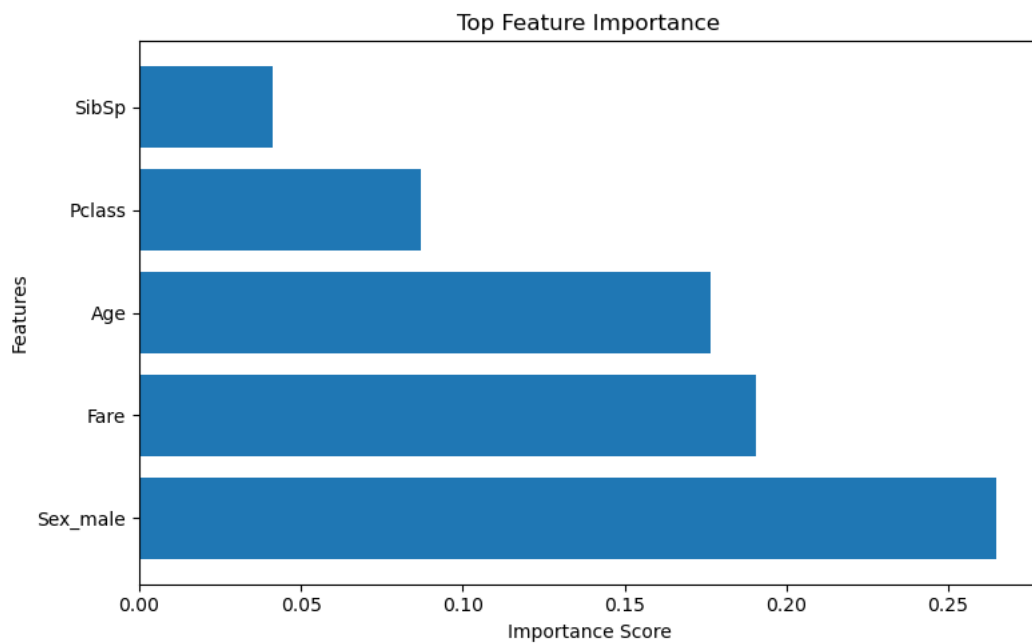
- Sex_male was the most influential feature.
- Fare was among the strongest predictors.
- Age contributed significantly to model decisions.

Recommendations

RECOMMENDATIONS

- Sex_male appears to be the strongest driver of predictions and should be monitored closely.
- The model demonstrates strong predictive performance, making the insights reasonably reliable.
- The most influential features should be tracked regularly, as changes in these variables are likely to have the largest effect on outcomes.

Feature Importance



Missing Values

